

# Chrystal Chern

Dynamics Lab / STAIRLab / Berkeley Discovery  
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*Dynamics researcher specializing in IMU-based motion sensing, system identification, and digital twins of physical systems, with experience spanning sensor instrumentation, standards-based experimental testing, and signal-level characterization through production-quality Python analysis tools.*

## Education

- 2019–2024 **PhD | Structural Engineering, Mechanics, and Materials, University of California, Berkeley,**  
GPA: 3.89/4.00.  
Thesis: *Digital Twin Framework for Vibration-Based Structural Health Monitoring*  
Advisor: Khalid M. Mosalam  
**Major Focus:** Structural mechanics and dynamics, linear and nonlinear structural analysis, finite element methods, performance-based engineering and design.  
**Minor Focus:** Artificial intelligence, machine learning, control systems, and scientific computing.
- 2019 **MS | Structural Engineering, Mechanics, and Materials, University of California, Berkeley.**  
Thesis: *Deep Learning for Transmission Tower Structural Health Monitoring in Small Datasets*
- 2016 **BS | Mechanical Engineering, Massachusetts Institute of Technology, Cambridge.**  
Mechanical systems, thermodynamics, and design.

## Research Experience

### University of California, Berkeley

- 2024–Present **Postdoctoral Researcher, Dynamics Lab & STAIRLab, Berkeley, CA.**  
**Dynamics Lab:**
- Lead instrumentation, signal processing, and inverse dynamic analysis of coupled human-object biomechanical systems using multi-node IMU sensor networks with strap-down integration and magnetic field mapping; led experiment design, sensor placement, and data collection.
  - Developed network analysis methodology and Python analysis code to identify the causal body-segment motions driving induced object motion (*Nonlinear Dynamics*, submitted 2026).
- STAIRLab (STructural Artificial Intelligence Research Lab):**
- Build finite element models to generate analytical benchmarks, validating system-identification methods across experimental, field, and FEA-simulated structural response data.
  - Evaluate system identification prediction residuals and modal clustering for anomaly detection and early-warning structural health monitoring under environmental and seismic loads.
  - Supervise graduate and undergraduate researchers in data-driven system dynamics.
- 2019–2024 **Graduate Student Researcher, STAIRLab & Pacific Earthquake Engineering Research Center (PEER), Berkeley, CA.**  
**Bridge Rapid Assessment Center for Extreme Events (BRACE2):**
- Developed California's production web platform for automated bridge health monitoring (Caltrans).
  - Performed system identification and finite element modeling of bridges and buildings.
- Structural ImageNet & Feature Engineering for Structural Health Monitoring:**
- Developed machine learning anomaly detection for bridge damage diagnosis within a human-machine collaboration framework (*Journal of Bridge Engineering*, 2024).
  - Developed deep learning image classification models for structural health monitoring.
  - Built feature engineering pipelines for damage classification.

## Massachusetts Institute of Technology

- 2015–2016 **Undergraduate Researcher**, *MIT Digital Structures Lab*, Cambridge, MA.  
Experimental characterization of fine-resolution anisotropic material properties of 3D printed materials via ASTM tensile testing.

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## Professional Engineering Experience

- 2016–2018 **Staff I – Building Technology**, *Simpson Gumpertz & Heger Inc.*, Washington, DC.
- Performed parametric energy, daylighting, and thermal finite element analysis (2D/3D) of curtain wall, cladding, and custom skylight assemblies.
  - Conducted infrared thermography surveys and analysis for non-destructive diagnostics.
  - Performed on-site water penetration and air infiltration testing of installed cladding assemblies per ASTM, AAMA, and ASHRAE standards, including instrumented pressure-chamber test setup, acceptance-criteria evaluation, and leak-path diagnosis.
  - Analyzed wind loads on components and cladding per structural code.
- 2014 **Engineering Intern**, *Dassault Systemes SOLIDWORKS Corp.*, Woodland Hills, CA.
- Investigated user-reported software issues across Computer Aided Design (CAD) and Finite Element Analysis (FEA) products for the SOLIDWORKS 2015 Beta program.
  - Contributed to the redesign of the engineering technical support knowledge base.

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## Technical & Computer Skills

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|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming (production)     | Python (package/module development), Bash, Tcl, HTML/CSS, JavaScript                                                                                                                                                                                                                           |
| Additional Languages & Tools | MATLAB, R, C, Rust                                                                                                                                                                                                                                                                             |
| Modeling & Analysis          | Vibrations and structural dynamics, differential equation modeling, linear algebra, numerical methods, finite element analysis of material and structural behavior (mechanical and thermal), machine learning anomaly detection                                                                |
| Test & Measurement           | Data acquisition (DAQ), sensor instrumentation, signal processing (FFT, Kalman filtering, strap-down integration, system identification), magnetic field mapping, ASTM standards-based testing (tensile, water penetration, air infiltration), infrared thermography, field acceptance testing |
| Languages                    | English (Native), Mandarin Chinese, Spanish                                                                                                                                                                                                                                                    |

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## Fellowships & Awards

- 2024–Present **UC Berkeley Future of Higher Education Postdoctoral Fellowship** (\$140k)
- 2020–2024 **NSF Graduate Research Fellowship (GRFP)** (\$160k)
- 2015 **MIT Priscilla King Gray Public Service Center Expedition Grant** (\$10k)
- 2012 **Intel Scholarship** (\$4k)

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## Publications

### Journal Articles

- 2026 **Chrystal Chern** and Oliver M. O'Reilly. "On Complex Modal Inverse Analysis: The Cautionary Tale of the Single Surviving Mode." *ASME Journal of Applied Mechanics*, 2026. <https://doi.org/10.1115/1.4072378>

- Submitted **Chrystal Chern**, Theresa E. Honein, and Oliver M. O'Reilly. "The Symphony of Gyration Producing Steady Motions of a Hula Hoop: Insights from a Network Analysis." *Nonlinear Dynamics*. Submitted May 2026.
- In Prep. **Chrystal Chern** and Khalid Mosalam. "Modal Clustering for Digital Twinning of Civil Infrastructure." *Journal of Engineering Mechanics*.
- In Prep. Naiqi Guo, **Chrystal Chern**, and Khalid Mosalam. "Detecting Inelasticity in Seismic Data with Time Response System Identification." *Mechanical Systems and Signal Processing*.
- 2024 Sifat Muin, **Chrystal Chern**, and Khalid Mosalam. "Human-Machine Collaboration Framework for Bridge Health Monitoring." *Journal of Bridge Engineering*, 29(7), 04024041. <https://doi.org/10.1061/JBENF2.BEENG-6587>

### Conference Proceedings

- 2025 **Chrystal Chern**, Claudio Perez, and Khalid Mosalam. "Structural Response Prediction from Learned System Realization Matrices." *11th International Conference on Experimental Vibration Analysis of Civil Engineering Structures (EVACES)*. [https://doi.org/10.1007/978-3-031-96110-6\\_72](https://doi.org/10.1007/978-3-031-96110-6_72)
- 2024 **Chrystal Chern** and Khalid Mosalam. "Cross-Sectional Study of Physics-Informed Bridge Health Identification." *International Association of Bridge Earthquake Engineering (IABEE) Fourth International Bridge Seismic Workshop (4IBSW)*.
- 2020 Sifat Muin, **Chrystal Chern**, and Khalid Mosalam. "Human-Machine Collaboration Framework for Bridge Health Monitoring." *SMIP20 Seminar on Utilization of Strong-Motion Data Proceedings*, pp. 100-127.

### Software

- 2024 **Chrystal Chern**, Claudio Perez, and Khalid Mosalam. mdof: 0.0.16-alpha, open-source package for system identification. Zenodo. <https://doi.org/10.5281/zenodo.11660201>

### Technical Reports & Dissertations

- 2024 **Chrystal Chern**. "Digital Twin Framework for Vibration-Based Structural Health Monitoring." *PhD Dissertation*, UC Berkeley.
- 2024 **Chrystal Chern**, Claudio Perez, and Khalid Mosalam. "BRACE2: Bridge rapid assessment center for extreme events, Phase I Final Report." *Caltrans Technical Report No. CA24-3703*.
- 2019 **Chrystal Chern**. "Deep Learning for Transmission Tower Structural Health Monitoring in Small Datasets." *MS Research Report*.

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## Teaching & Mentorship

- Spring 2026 **Lecturer (Instructor of Record)**, L&S 110 – *Brilliance of Berkeley* (UC Berkeley). Advise students, prepare course materials, grade, facilitate discussions, and coordinate guest speakers; supervise student readers.
- 2024–2025 **Teaching Assistant**, MAS-E ENGIN 235-A – *Python for Engineers* (UC Berkeley). Authored programming laboratory assignments, developed autograder scripts, and reviewed lectures and quizzes.
- Dec 2024 **Guest Lecturer**, CE 249 – *Experimental Methods in Structural Engineering* (UC Berkeley). Delivered a lecture on system identification of vibration data using computer vision.

- 2019, **Graduate Student Instructor, E7 – Computer Programming for Scientists and Engineers (UC Berkeley)**.  
Delivered discussion lectures, worksheets, and laboratory assignments; led exam review sessions and graded exams.
- 2025 **Lead Organizer & Founder, SEMM Graduate Program Primer Boot Camp, UC Berkeley**.  
Founded and taught a primer on core theory and field-specific conventions for incoming graduate students; organized 7 doctoral co-instructors.

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## Service & Leadership

- 2024–Present **Program Strategist**, Berkeley Discovery Initiative, UC Berkeley – Co-developed an interdisciplinary research mentorship program (8 research hubs, 30+ graduate fellows, 170 undergraduates); led survey and focus-group assessment and co-secured \$170k in continuing funding.
- 2022–Present **Technical Reviewer** for *Resilient Cities and Structures*, *Computers and Structures*, and *Computer-Aided Civil and Infrastructure Engineering*.
- 2024–Present **Research Mentor**, UC Berkeley SEMM & CEE Scholars Programs – Designed and supervised structural health monitoring and sensor projects for 5 undergraduate researchers.
- 2022–2024 **Founder & President**, UC Berkeley SEMM-DSR (Doctoral Students and Researchers Organization).
- 2014–2015 **Malawi Project Manager & Fundraising Chair**, MIT Engineers Without Borders.

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## References

### Oliver M. O'Reilly

Postdoctoral Research Supervisor

Distinguished Professor of Mechanical Engineering, UC Berkeley

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### Khalid M. Mosalam

PhD Thesis Advisor & Postdoctoral Supervisor

Taisei Professor of Civil Engineering & Director, PEER Center, UC Berkeley

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### Leslie R. Harlson

Postdoctoral Supervisor

Executive Director, Berkeley Discovery, UC Berkeley

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### Alessandra Lanzara

Postdoctoral Supervisor, Berkeley Discovery

Charles Kittel Chair in Physics & Faculty Director, Kavli ENSI, UC Berkeley

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